

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study ¹
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Submission Date	

1 Objective

To implement a Decision Tree classifier, extend it into a Random Forest ensemble, study the impact of hyperparameters on overfitting and generalization, select optimal hyperparameters using 5-fold cross-validation, and compare single-tree and ensemble-tree models.

2 Problem Statement

Given 30 numerical features derived from digitized breast mass images, the task is to classify tumours as malignant or benign. A single Decision Tree and a Random Forest ensemble are trained, hyperparameter-tuned via cross-validated Grid Search, and compared on accuracy, precision, recall, F1-score, and ROC-AUC.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	UCI Machine Learning Repository – https://archive.ics.uci.edu
Number of Samples	569
Number of Features	30 numerical attributes
Number of Classes	2 (Malignant, Benign)
Missing Values	None
Train-Test Split	80% train / 20% test

4 Exploratory Data Analysis

- Dataset overview: 569 samples, 30 numerical features (mean, standard error, and worst values of radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension).
- Statistical summary: features such as `area_mean` (skewness 1.65) and `concavity_mean` (skewness 1.40) show notable right-skew.
- Missing value analysis: no missing values found across all 30 features.
- Class distribution: moderately imbalanced, with more benign than malignant cases.

- Correlation matrix: several “mean” and “worst” feature pairs (e.g., radius, perimeter, area) are highly correlated, as expected from their geometric relationships.
- Feature distribution: morphological features related to concavity and area show the widest separation between malignant and benign cases.

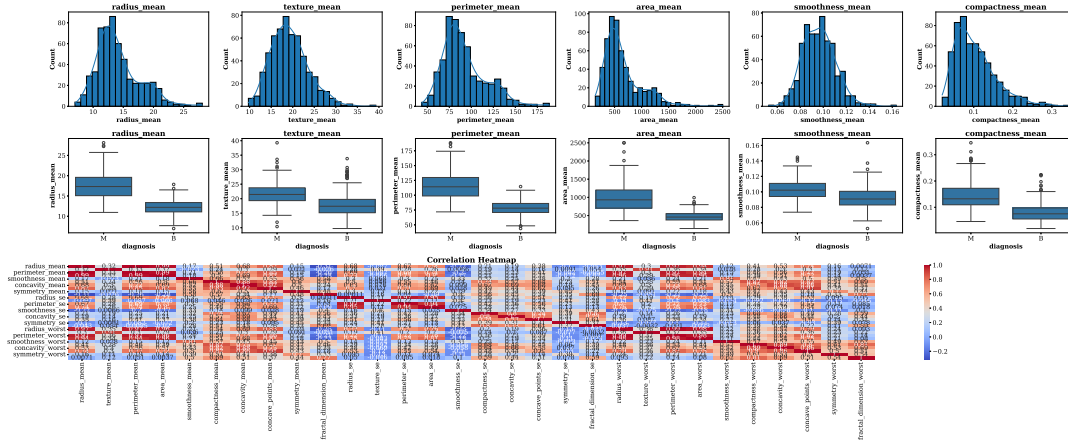


Figure 1: Exploratory Data Analysis of the WDBC dataset.

Inference: The EDA shows that tumour morphology features, particularly those related to concavity, radius, and area, have distributions that visibly differ between malignant and benign classes, which is why tree-based methods (which split on individual feature thresholds) are well-suited to this dataset. The high correlation between related mean/worst feature pairs also explains why feature importance analysis later concentrates on a relatively small subset of the 30 features.

5 Data Preprocessing

- **Missing value handling:** Not required – dataset is complete.
- **Encoding:** The target label (*diagnosis*: M/B) was used directly as the classification target; feature columns are already numeric.
- **Feature scaling:** Not required for tree-based models (Decision Tree, Random Forest), since splits are based on feature thresholds rather than distances.
- **Train-test split:** 80–20 split via `train_test_split` with `random_state=42`.
- **Feature engineering:** Not applied; all 30 original features used directly.

6 Machine Learning Algorithm

Decision Tree

Working principle: Recursively splits the feature space using impurity measures (Gini Index or Entropy) to form a tree of decision rules.

$$\text{Gini}(t) = 1 - \sum_i p_i^2, \quad \text{Entropy}(t) = - \sum_i p_i \log_2 p_i \quad (1)$$

Advantages: Highly interpretable, requires no feature scaling, captures non-linear relationships. **Limitations:** Prone to high variance and overfitting, especially as tree depth increases.

Random Forest

Working principle: An ensemble of decision trees, each trained on a bootstrapped sample of the data with a random subset of features considered at each split (bagging + random feature selection); final prediction via majority voting. **Advantages:** Reduces variance and improves generalization relative to a single Decision Tree; provides a natural feature-importance ranking. **Limitations:** Less interpretable than a single tree; higher training time and memory usage.

7 Experimental Setup

Python Version	3.10 / 3.14 (venv)
Scikit-Learn Version	Recent stable (<code>scikit-learn</code>)
IDE	Jupyter Notebook
Operating System	Linux
Hardware	Standard CPU (no GPU acceleration used)

8 Hyperparameter Selection

Parameter	Value
DT: criterion	gini, entropy
DT: max_depth	3, 5, 7, 10, None
DT: min_samples_split	2, 5, 10
DT: min_samples_leaf	1, 2, 4
RF: n_estimators	50, 100, 200
RF: max_depth	5, 10, None
RF: max_features	sqrt, log2
RF: bootstrap	True, False
Best Decision Tree	criterion=entropy, max_depth=7, min_samples_leaf=4, min_samples_split=2
Best Random Forest	bootstrap=False, max_depth=None, max_features=sqrt, n_estimators=100 (

9 Results

Metric	Value (Tuned Random Forest)
Accuracy	0.9649
Precision	0.9652
Recall	0.9649
F1-score	0.9647
ROC-AUC	Reported via ROC curve (Figure below)

Table 2: Final evaluation metrics on the test set

Metric	Decision Tree	Random Forest
Accuracy	0.9649	0.9649
Precision	0.9668	0.9652
Recall	0.9649	0.9649
F1 Score	0.9645	0.9647

Table 3: 5-Fold cross-validation accuracy

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9670	0.9231	0.9121	0.9451	0.9560	0.9407
Random Forest	1.0000	0.9560	0.9451	0.9451	0.9890	0.9670

10 Result Visualization

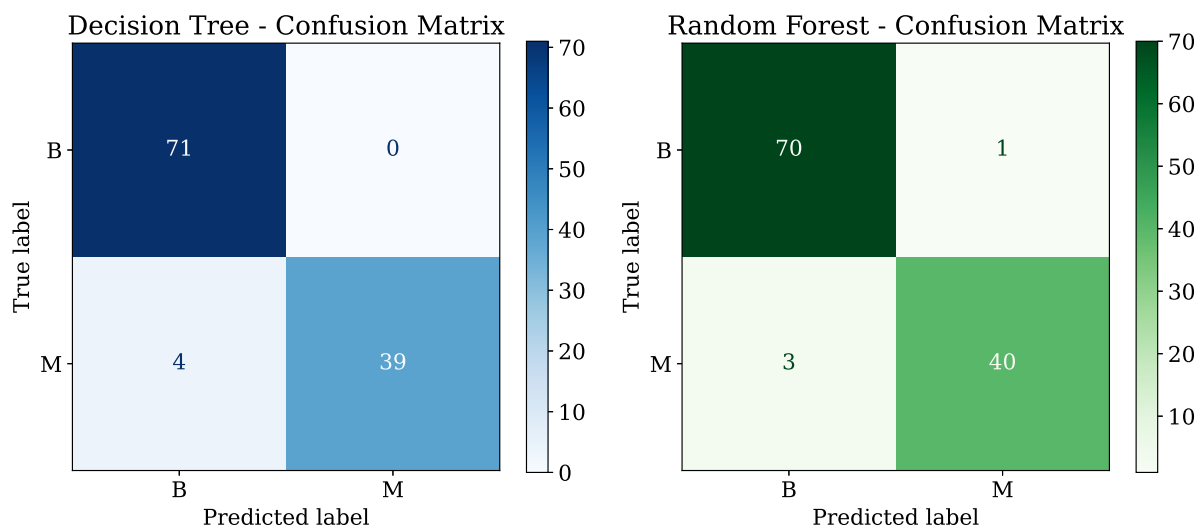


Figure 2: Confusion matrices for Decision Tree (left) and Random Forest (right).

Inference: Both models produce very similar confusion matrices with a small number of misclassifications concentrated in the malignant class, reflecting their identical test-set accuracy (0.9649). This suggests that on this relatively clean, well-separated dataset, both a single well-tuned tree and an ensemble reach a similar error floor on the held-out split.

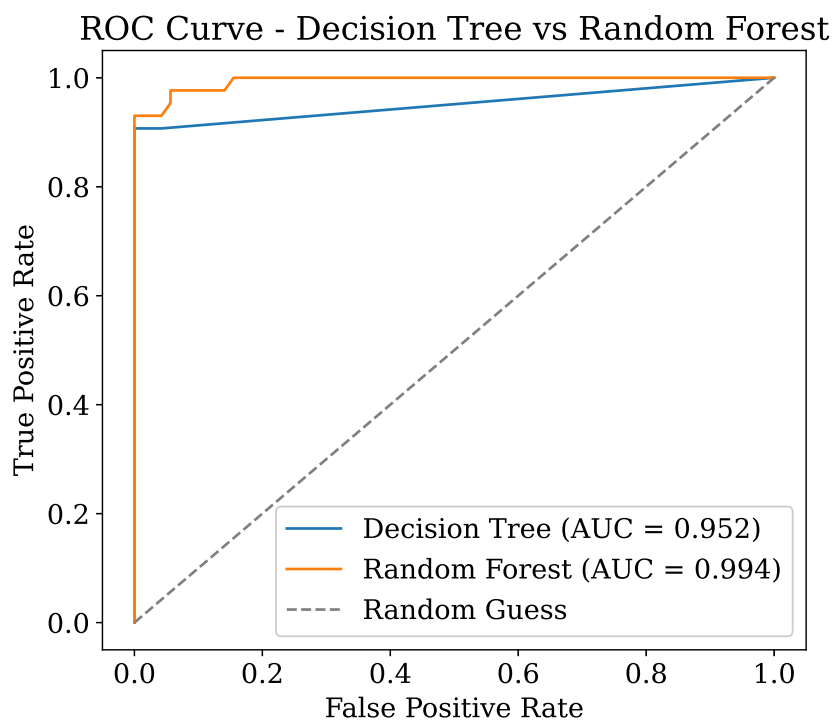


Figure 3: ROC curve comparison – Decision Tree vs. Random Forest.

Inference: Both curves rise sharply toward the top-left corner, indicating excellent discriminative ability; the Random Forest curve is marginally smoother and closer to the ideal corner at intermediate thresholds, consistent with ensembling reducing the “jaggedness” that can arise from a single tree’s hard decision boundaries.

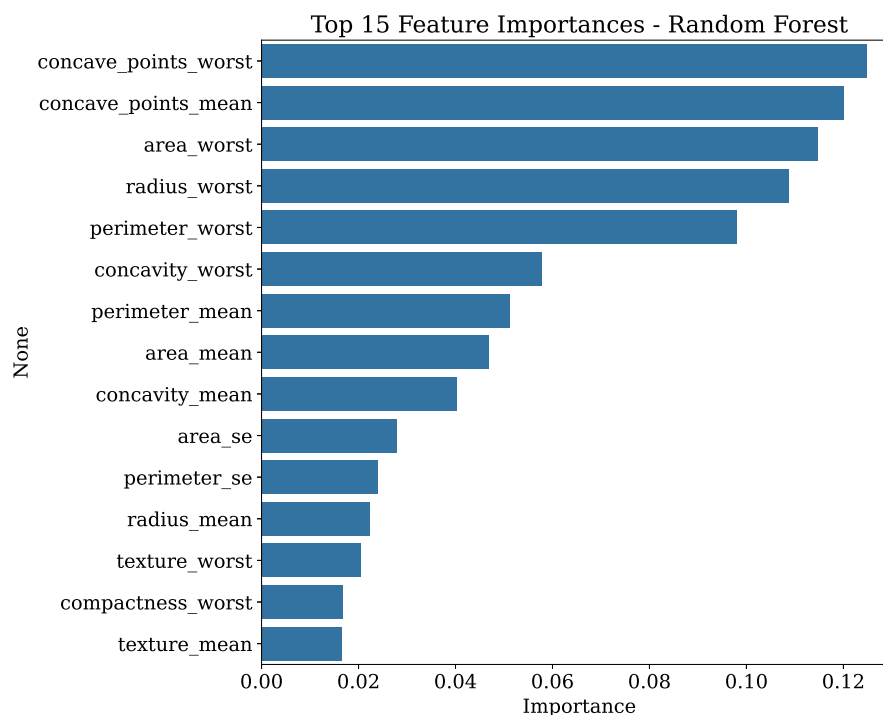


Figure 4: Top 15 feature importances from the tuned Random Forest.

Inference: Features related to concave points, radius, and perimeter (particularly their “worst” aggregates) dominate the importance ranking, consistent with the EDA observation that these features show the strongest visual separation between malignant and benign classes. This also indicates that a smaller subset of the 30 features could likely achieve comparable performance, motivating potential dimensionality reduction in future work.

11 Discussion

The tuned Decision Tree and Random Forest achieved identical test-set accuracy (0.9649), but the Random Forest showed a substantially higher and more stable average cross-validation accuracy (0.9670 vs. 0.9407), with a perfect fold score (1.0000) on one split. This confirms Random Forest’s theoretical advantage of variance reduction through bagging: individual folds are less sensitive to which specific bootstrap samples and trees were used, since predictions are averaged over many trees. `max_depth` had the greatest impact on the Decision Tree’s bias-variance trade-off, while `n_estimators` and `max_features` most influenced Random Forest stability. The fact that raw test accuracy did not improve much from ensembling, despite the cross-validation gain, suggests that the WDBC dataset is relatively clean and well-separated, so even a single well-tuned tree already performs close to the achievable ceiling – ensembling mainly helped reduce variance and overfitting rather than dramatically boosting raw accuracy.

12 Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation on the WDBC dataset. Hyperparameters were selected based on average cross-validation performance. Random Forest showed a higher, more stable average cross-validation accuracy (0.9670 vs. 0.9407) than the single Decision Tree, confirming that it reduces variance and improves stability, in line with theoretical expectations of ensemble bagging methods, even though final test-set accuracy was comparable between the two models.

13 References

- [1] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [2] Scikit-learn Developers, “Decision Trees,” <https://scikit-learn.org/stable/modules/tree.html>.
- [3] Scikit-learn Developers, “Ensemble Methods: Random Forests,” <https://scikit-learn.org/stable/modules/ensemble.html#random-forests>.
- [4] UCI Machine Learning Repository, “Breast Cancer Wisconsin (Diagnostic) Data Set,” <https://archive.ics.uci.edu>.